2.13 Maclaurin Series: Limits

Worked Solutions

1 a)

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots .$$

$$\begin{aligned}\lim_{x\to 0} \frac{\ln(1+x)}{x} &= \lim_{x\to 0} \frac{x - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \cdots}{x} \\ &= \lim_{x\to 0} \left(1 - \frac{x}{2} + \frac{x^2}{3} - \cdots\right) = 1.\end{aligned}$$

1 b)

$$e^x = 1 + x + \frac{x^2}{2} + \cdots .$$

$$\begin{aligned}\lim_{x\to 0} \frac{e^x - 1}{2x} &= \lim_{x\to 0} \frac{x + \frac{x^2}{2} + \cdots}{2x} \\ &= \lim_{x\to 0} \left(\frac{1}{2} + \frac{x}{4} + \cdots\right) = \frac{1}{2}.\end{aligned}$$

1 c)

$$\cos x + \sin x = 1 - \frac{1}{2}x^2 + \cdots + x - \frac{1}{6}x^3 + \cdots , \qquad e^x = 1 + x + \frac{1}{2}x^2 + \cdots .$$

$$\lim_{x\to 0} \frac{\cos x + \sin x}{e^x} = \lim_{x\to 0} \frac{1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots}{1 + x + \frac{1}{2}x^2 + \cdots} = 1.$$

2 a)

$$\ln(1+x^2) = x^2 - \frac{x^4}{2} + \frac{x^6}{3} - \cdots , \quad \lim_{x\to 0} \frac{\ln(1+x^2)}{x^2} = 1.$$

2 b)

$$(1+3x)^{-2} = 1 - 6x + 27x^2 - \cdots , \quad \lim_{x\to 0} \frac{-1 + (1+3x)^{-2}}{x} = \lim_{x\to 0}(-6 + 27x - \cdots) = -6.$$

2 c)

$$\sin 4x = 4x - \frac{1}{6}(4x)^3 + \cdots = 4x - \frac{32}{2}x^3 + \cdots , \quad \lim_{x\to 0} \frac{\sin 4x}{x} = \lim_{x\to 0} \left(4 - \frac{32}{2}x^2 + \cdots\right) = 4.$$

2 d)

$$\cos 3x = 1 - \frac{9}{2}x^2 + \cdots, \quad \lim_{x\to 0} \frac{x^2}{1-\cos 3x} = \lim_{x\to 0} \frac{x^2}{\frac{9}{2}x^2 + \cdots} = \frac{2}{9}.$$

2 e)

$$e^{-2x} = 1 - 2x + 2x^2 + \cdots, \quad \lim_{x\to 0} \frac{1-e^{-2x}}{x} = \lim_{x\to 0}(2 - 2x + \cdots) = 2.$$

2 f)

$$e^{3x} = 1 + 3x + \frac{9}{2}x^2 + \cdots, \quad e^x = 1 + x + \frac{1}{2}x^2 + \cdots.$$

$$\lim_{x\to 0} \frac{e^{3x} - e^x}{x} = \lim_{x\to 0}(2 + 4x + \cdots) = 2.$$

2 g)

$$e^{2x} = 1 + 2x + 2x^2 + \cdots, \quad \lim_{x\to 0} \frac{e^{2x}-2}{2e^{2x}+1} = \lim_{x\to 0} \frac{-1 + 2x + 2x^2 + \cdots}{3 + 4x + 4x^2 + \cdots} = -\frac{1}{3}.$$

3)

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \qquad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots.$$

$$\lim_{x\to 0} \frac{\ln(1+x) - x}{1-\cos x} = \lim_{x\to 0} \frac{-\frac{x^2}{2} + \frac{x^3}{3} - \cdots}{\frac{x^2}{2} - \frac{x^4}{24} + \cdots}$$

$$= \lim_{x\to 0} \frac{-\frac{1}{2} + \frac{x}{3} - \cdots}{\frac{1}{2} - \frac{x^2}{24} + \cdots} = -1.$$

4 a)

$$\sqrt{x^2+5x} = x\sqrt{1+\frac{5}{x}} = x\left(1+\frac{5}{x}\right)^{1/2} = x\left[1 + \frac{5}{2x} - \frac{25}{8x^2} + \cdots\right] = x + \frac{5}{2} - \frac{25}{8x} + \cdots.$$

$$\lim_{x\to\infty}\left[x - \sqrt{x^2+5x}\right] \approx \lim_{x\to\infty}\left[-\frac{5}{2} + \frac{25}{8x}\right] = -\frac{5}{2}.$$

4 b)

$$\sqrt[3]{x^3+7} = x\left(1+\frac{7}{x^3}\right)^{1/3} = x\left[1 + \frac{7}{3x^3} - \frac{49}{9x^6} + \cdots\right] = x + \frac{7}{3x^2} - \frac{49}{9x^5} + \cdots.$$

$$\lim_{x\to\infty}\left[\sqrt[3]{x^3+7} - x\right] \approx \lim_{x\to\infty}\left[\frac{7}{3x^2} - \frac{49}{9x^5}\right] = 0.$$

4 c)

$$\begin{aligned}\sqrt{x^2-8x+3} &= x\left(1-\frac{8}{x}+\frac{3}{x^2}\right)^{1/2}\\ &= x\left[1+\frac{1}{2}\left(-\frac{8}{x}+\frac{3}{x^2}\right)-\frac{1}{8}\left(-\frac{8}{x}+\frac{3}{x^2}\right)^2+\cdots\right]\\ &= x\left[1-\frac{4}{x}+\frac{3}{2x^2}-\frac{8}{x^2}+\frac{6}{x^3}-\frac{9}{8x^4}+\cdots\right]\\ &= x-4-\frac{13}{2x}+\cdots.\end{aligned}$$

$$\lim_{x\to\infty}\left[x-\sqrt{x^2-8x+3}\right] \approx \lim_{x\to\infty}\left[4+\frac{23}{2x}+\cdots\right]=4.$$

5 a)

$$\begin{aligned}2x\cos 3x+\sin 2x &= 2x\left(1-\frac{(3x)^2}{2}+\frac{(3x)^4}{24}+\cdots\right)+2x-\frac{(2x)^3}{6}+\frac{(2x)^5}{120}+\cdots\\ &\approx 2x-9x^3+\frac{27}{4}x^5+2x-\frac{4}{3}x^3+\frac{4}{15}x^5\\ &= 4x-\frac{31}{3}x^3+\frac{421}{60}x^5.\end{aligned}$$

5 b)

$$\begin{aligned}\lim_{x\to 0}\frac{-4x+2x\cos 3x+\sin 2x}{x^3} &\approx \lim_{x\to 0}\frac{-4x+4x-\frac{31}{3}x^3+\frac{421}{60}x^5}{x^3}\\ &= \lim_{x\to 0}\left(-\frac{31}{3}+\frac{421}{60}x^2\right)=-\frac{31}{3}.\end{aligned}$$

6 a)

$$\sin 3x = 3x-\frac{(3x)^3}{3!}+\frac{(3x)^5}{5!}+\cdots\approx 3x-\frac{9}{2}x^3+\frac{81}{40}x^5.$$

6 b)

$$\cos 4x = 1-\frac{(4x)^2}{2!}+\frac{(4x)^4}{4!}+\cdots\approx 1-8x^2+\frac{32}{3}x^4.$$

6 c)

$$\begin{aligned}\lim_{x\to 0}\frac{3x\cos 4x-\sin 3x}{2x^3} &\approx \lim_{x\to 0}\frac{3x(1-8x^2+\frac{32}{3}x^4)-(3x-\frac{9}{2}x^3+\frac{81}{40}x^5)}{2x^3}\\ &= \lim_{x\to 0}\frac{-\frac{39}{2}x^3+\frac{1199}{40}x^5}{2x^3}=-\frac{39}{4}.\end{aligned}$$

7 a)

$$(1+3x)^{-1/3} = 1 - x + 2x^2 - \frac{14}{3}x^3 + \cdots, \qquad e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots.$$

$$\begin{aligned} f(x) &= e^x(1+3x)^{-1/3} \\ &= (1 + x + \tfrac{1}{2}x^2 + \tfrac{1}{6}x^3)(1 - x + 2x^2 - \tfrac{14}{3}x^3) + \cdots \\ &= 1 - x + 2x^2 - \tfrac{14}{3}x^3 + x - x^2 + 2x^3 + \tfrac{1}{2}x^2 - \tfrac{1}{2}x^3 + \tfrac{1}{6}x^3 + \cdots \\ &= 1 + \frac{3}{2}x^2 - 3x^3 + \cdots. \end{aligned}$$

7 b)

$$g(x) = \ln(1+2x) = 2x - 2x^2 + \cdots.$$

$$\lim_{x\to 0} \frac{1 - f(x)}{x g(x)} \approx \lim_{x\to 0} \frac{-\frac{3}{2}x^2 + 3x^3}{2x^2 - 2x^3} = -\frac{3}{4}.$$

8 a)

$$\ln(1+3x) = 3x - \frac{(3x)^2}{2} + \frac{(3x)^3}{3} - \frac{(3x)^4}{4} + \cdots \approx 3x - \frac{9}{2}x^2 + 9x^3 - \frac{81}{4}x^4.$$

8 b)

$$\begin{aligned} \ln[(1+3x)(1-3x)] &= \ln(1+3x) + \ln(1-3x) \\ &\approx \left(3x - \frac{9}{2}x^2 + 9x^3 - \frac{81}{4}x^4\right) + \left(-3x - \frac{9}{2}x^2 - 9x^3 - \frac{81}{4}x^4\right) \\ &= -9x^2 - \frac{81}{2}x^4. \end{aligned}$$

8 c)

$$\lim_{x\to 0} \frac{\ln[(1+3x)(1-3x)]}{4x^2} \approx \lim_{x\to 0} \left(-\frac{9}{4} - \frac{81}{8}x^2\right) = -\frac{9}{4}.$$

9 a)

$$e^{5x^2} = 1 + 5x^2 + \frac{1}{2}(5x^2)^2 + \cdots \approx 1 + 5x^2 + \frac{25}{2}x^4.$$

9 b)

$$\lim_{x\to 0} \frac{e^{5x^2}}{k - \cos x} \approx \lim_{x\to 0} \frac{1 + 5x^2 + \frac{25}{2}x^4}{k - (1 - \frac{1}{2}x^2 + \frac{1}{24}x^4)} = \frac{1}{k-1} = -6, \quad \therefore k = \frac{5}{6}.$$

10 a)

$$\frac{6(1-x)}{(x+1)(x+3)} \equiv \frac{A}{x+1} + \frac{B}{x+3}, \quad 6-6x = A(x+3) + B(x+1).$$
$$x=-1:\ 12 = 2A \Rightarrow A = 6, \qquad x=-3:\ 24 = -2B \Rightarrow B = -12.$$
$$\frac{6(1-x)}{(x+1)(x+3)} = \frac{6}{x+1} - \frac{12}{x+3}.$$

10 b)

$$\begin{aligned} f(x) &= 6(1+x)^{-1} - 12(3+x)^{-1} = 6(1+x)^{-1} - 4(1+\tfrac{1}{3}x)^{-1} \\ &\approx 6(1-x+x^2) - 4(1-\tfrac{1}{3}x+\tfrac{1}{9}x^2) \\ &= 2 - \frac{14}{3}x + \frac{50}{9}x^2. \end{aligned}$$

10 c)

$$\lim_{x\to 0} f(x) \approx \lim_{x\to 0}\left(2 - \frac{14}{3}x + \frac{50}{9}x^2\right) = 2.$$

11 a)

$$\frac{3-x}{(x+1)(x-5)} \equiv \frac{A}{x+1} + \frac{B}{x-5}, \quad 3-x = A(x-5) + B(x+1).$$
$$x=-1:\ 4 = -6A \Rightarrow A = -\frac{2}{3}, \qquad x=5:\ -2 = 6B \Rightarrow B = -\frac{1}{3}.$$
$$\frac{3-x}{(x+1)(x-5)} = -\frac{2}{3(x+1)} - \frac{1}{3(x-5)}.$$

11 b)

$$\begin{aligned} f(x) &= -\frac{2}{3}(1+x)^{-1} - \frac{1}{3}(x-5)^{-1} \\ &\approx -\frac{2}{3}(1-x+x^2) - \frac{1}{3}\left(-\frac{1}{5} - \frac{x}{25} - \frac{x^2}{125}\right) \\ &= -\frac{3}{5} + \frac{17}{25}x - \frac{83}{125}x^2. \end{aligned}$$

11 c)

$$\lim_{x\to 0} \frac{3-x}{2(x+1)(x-5)} = \lim_{x\to 0} \frac{1}{2} f(x) = -\frac{3}{10}.$$

12 a)

$$\frac{4}{(x-1)(x^2+2)} \equiv \frac{A}{x-1} + \frac{Bx+C}{x^2+2}, \quad 4 = A(x^2+2) + (Bx+C)(x-1).$$

$$x = 1: \quad 4 = 3A \Rightarrow A = \frac{4}{3};$$

$$x = 0: \quad 4 = 2A - C \Rightarrow C = -\frac{4}{3};$$

$$x = -1: \quad 4 = 3A + 2B - 2C \Rightarrow B = -\frac{4}{3}.$$

$$\frac{4}{(x-1)(x^2+2)} = \frac{4}{3(x-1)} - \frac{4x+4}{3(x^2+2)}.$$

12 b)

$$\begin{aligned} f(x) &= \frac{4}{3}(x-1)^{-1} - \frac{4}{3}(x+1)(2+x^2)^{-1} \\ &= -\frac{4}{3}(1-x)^{-1} - \frac{2}{3}(x+1)(1+\tfrac{1}{2}x^2)^{-1} \\ &\approx -\frac{4}{3}(1+x+x^2) - \frac{2}{3}(x+1)(1-\tfrac{1}{2}x^2) \\ &= -2 - 2x - x^2. \end{aligned}$$

12 c)

$$\lim_{x\to 0} f(x) \approx \lim_{x\to 0}(-2-2x-x^2) = -2.$$

13 a)

$$\cos 2x = 1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \frac{(2x)^6}{6!} + \cdots \approx 1 - 2x^2 + \frac{2}{3}x^4 - \frac{4}{45}x^6.$$

13 b)

$$\begin{aligned} \lim_{x\to 0} \frac{1-\cos 2x}{x\ln(1+\frac{1}{2}x)} &\approx \lim_{x\to 0} \frac{2x^2 - \frac{2}{3}x^4 + \frac{4}{45}x^6}{x\ln(1+\frac{1}{2}x)} \\ &= \lim_{x\to 0} \frac{2x - \frac{2}{3}x^3 + \frac{4}{45}x^5}{\ln(1+\frac{1}{2}x)}. \end{aligned}$$

Using L'Hôpital's rule,

$$\lim_{x\to 0} \frac{2x - \frac{2}{3}x^3 + \frac{4}{45}x^5}{\ln(1+\frac{1}{2}x)} = \lim_{x\to 0} \frac{2 - 2x^2 + \frac{4}{9}x^4}{\frac{1}{2(1+\frac{1}{2}x)}} = 4.$$

Alternatively,

$$\begin{aligned}\lim_{x\to 0}\frac{1-\cos 2x}{x\ln(1+\frac{1}{2}x)} &\approx \lim_{x\to 0}\frac{2x^2-\frac{2}{3}x^4+\frac{4}{45}x^6}{x(\frac{1}{2}x-\frac{1}{8}x^2+\cdots)}\\ &= \lim_{x\to 0}\frac{2x^2-\frac{2}{3}x^4+\frac{4}{45}x^6}{\frac{1}{2}x^2-\frac{1}{8}x^3}\\ &\xrightarrow{\text{L'Hôpital}} \lim_{x\to 0}\frac{4x-\frac{8}{3}x^3+\frac{8}{15}x^5}{x-\frac{3}{8}x^2}\\ &= \lim_{x\to 0}\frac{4-8x^2+\frac{8}{3}x^4}{1-\frac{3}{4}x} = 4.\end{aligned}$$

14 a)

$$e^{-3x} = 1-3x+\frac{1}{2}(-3x)^2+\frac{1}{6}(-3x)^3+\cdots \approx 1-3x+\frac{9}{2}x^2-\frac{9}{2}x^3.$$

14 b)

$$\begin{aligned}\lim_{x\to 0}\frac{1-e^{-3x}}{\ln(1+kx)} &\approx \lim_{x\to 0}\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{\ln(1+kx)}\\ &\xrightarrow{\text{L'Hôpital}} \lim_{x\to 0}\frac{3-9x+\frac{27}{2}x^2}{k/(1+kx)}.\end{aligned}$$

So that $\frac{3}{k}=\frac{1}{2}$, hence $k=6$. Alternatively,

$$\begin{aligned}\lim_{x\to 0}\frac{1-e^{-3x}}{\ln(1+kx)} &\approx \lim_{x\to 0}\frac{3x-\frac{9}{2}x^2+\frac{9}{2}x^3}{kx-\frac{k^2}{2}x^2+\cdots}\\ &\xrightarrow{\text{L'Hôpital}} \lim_{x\to 0}\frac{3-9x+\frac{27}{2}x^2}{k-k^2x+\cdots}.\end{aligned}$$

So that So that $\frac{3}{k}=\frac{1}{2}$, hence $k=6$.

15 a)

$$\begin{aligned}\ln(e^{2x}\sin x) &= \ln(e^{2x})+\ln(\sin x) = 2x+\ln(\sin x),\\ \sin x &= x-\frac{1}{6}x^3+\frac{1}{120}x^5-\cdots,\\ \ln(\sin x) &= \ln\left[x\left(1-\frac{1}{6}x^2+\frac{1}{120}x^4-\cdots\right)\right]\\ &= \ln x+\ln\left(1-\frac{1}{6}x^2+\frac{1}{120}x^4-\cdots\right)\\ &= \ln x+\left(-\frac{1}{6}x^2+\frac{1}{120}x^4\right)-\frac{1}{2}\left(-\frac{1}{6}x^2+\frac{1}{120}x^4\right)^2+\cdots.\\ &\therefore\quad \ln(e^{2x}\sin x)\approx \ln x+2x-\frac{1}{6}x^2.\end{aligned}$$

15 b)

$$\ln(1+kx) = kx-\frac{1}{2}(kx)^2+\cdots \approx kx-\frac{k^2}{2}x^2.$$

15 c)

$$\lim_{x\to 0}\frac{\ln(e^{2x}\sin x)-\ln x}{\ln(1+kx)} \approx \lim_{x\to 0}\frac{2x-\frac{1}{6}x^2}{kx-\frac{k^2}{2}x^2}$$
$$= \lim_{x\to 0}\frac{2-\frac{1}{6}x}{k-\frac{k^2}{2}x} = \frac{2}{k}.$$

So it exists for any $k \neq 0$, and the limit is $\frac{2}{k}$.

16 a)

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots .$$

16 b)

$$\sum_{r=1}^{\infty}\frac{4-r}{r!} = 4\sum_{r=1}^{\infty}\frac{1}{r!} - \sum_{r=1}^{\infty}\frac{r}{r!}.$$

Since $e^x - 1 = x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$, $\sum_{r=1}^{\infty}\frac{1}{r!} = e - 1$. Also $\frac{r}{r!} = \frac{1}{(r-1)!}$, so $\sum_{r=1}^{\infty}\frac{r}{r!} = \sum_{r=1}^{\infty}\frac{1}{(r-1)!} = e$. Therefore

$$\sum_{r=1}^{\infty}\frac{4-r}{r!} = 4(e-1) - e = 3e - 4.$$

16 c)

$$4(e^x-1) - xe^x = 4\sum_{r=1}^{\infty}\frac{x^r}{r!} - x\sum_{r=0}^{\infty}\frac{x^r}{r!}$$
$$= \sum_{r=1}^{\infty}\frac{4x^r}{r!} - \sum_{r=1}^{\infty}\frac{x^r}{(r-1)!}$$
$$= \sum_{r=1}^{\infty}\left(\frac{4}{r!} - \frac{1}{(r-1)!}\right)x^r$$
$$= \sum_{r=1}^{\infty}\frac{4-r}{r!}x^r.$$

$$\frac{4(e^x-1)-xe^x}{x} = \sum_{r=1}^{\infty}\frac{4-r}{r!}x^{r-1}, \quad \lim_{x\to 0}\frac{4(e^x-1)-xe^x}{x} = \frac{4-1}{1!} = 3.$$

17 a)

$$f(x) = \sqrt{4-x}, \qquad f(0) = 2,$$
$$f'(x) = -\frac{1}{2}(4-x)^{-1/2}, \quad f'(0) = -\frac{1}{4},$$
$$f''(x) = -\frac{1}{4}(4-x)^{-3/2}, \quad f''(0) = -\frac{1}{32}.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots , \quad \sqrt{4-x} = 2 - \frac{1}{4}x - \frac{1}{64}x^2 + \cdots .$$

$$\therefore \quad \sqrt{4-x} \approx 2 - \frac{1}{4}x - \frac{1}{64}x^2.$$

17 b)

$$\lim_{x\to 0} \frac{\sqrt{4-x}}{1+x} \approx \lim_{x\to 0} \frac{2 - \frac{1}{4}x - \frac{1}{64}x^2}{1+x} = 2.$$

18 a)

$$f(x) = \sqrt{2+e^x}, \quad f(0) = \sqrt{3}, \quad f'(x) = \frac{1}{2}e^x(2+e^x)^{-1/2}, \quad f'(0) = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$\sqrt{2+e^x} = \sqrt{3} + \frac{\sqrt{3}}{6}x + \cdots, \quad \therefore \quad \sqrt{2+e^x} \approx \sqrt{3} + \frac{\sqrt{3}}{6}x.$$

18 b)

$$\lim_{x\to 0} \frac{\sqrt{2+e^x} - 1}{\cos 2x} \approx \lim_{x\to 0} \frac{(\sqrt{3} + \frac{\sqrt{3}}{6}x) - 1}{1 - 2x^2} = \sqrt{3} - 1.$$

19 a)

$$f(x) = \sqrt{1+\tan x}, \quad f(0) = 1, \quad f'(x) = \frac{1}{2}\sec^2 x(1+\tan x)^{-1/2}, \quad f'(0) = \frac{1}{2}.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$\sqrt{1+\tan x} = 1 + \frac{1}{2}x + \cdots, \quad \therefore \quad \sqrt{1+\tan x} \approx 1 + \frac{1}{2}x.$$

19 b)

$$\lim_{x\to 0} \frac{\sqrt{1+\tan x} - 1}{\sin kx} \approx \lim_{x\to 0} \frac{\frac{1}{2}x}{kx - \frac{k^3}{6}x^3} = \frac{1}{2k} = \frac{1}{4}, \quad \therefore k = 2.$$

20 a)

$$f(x) = (1+\sin x)^{3/2}, \quad f(0) = 1, \quad f'(x) = \frac{3}{2}\cos x(1+\sin x)^{1/2}, \quad f'(0) = \frac{3}{2}.$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$(1+\sin x)^{3/2} = 1 + \frac{3}{2}x + \cdots, \quad \therefore \quad (1+\sin x)^{3/2} \approx 1 + \frac{3}{2}x.$$

20 b)

$$\lim_{x\to 0}\frac{(1+\sin x)^{3/2}-1}{\sin kx}\approx\lim_{x\to 0}\frac{\frac{3}{2}x}{kx-\frac{k^3}{6}x^3}=\frac{3}{2k}=\frac{3}{8},\quad\therefore k=4.$$

21 a)

$$\begin{aligned}
f(x)&=\cosh x \text{ so } f(0)=1,\\
f'(x)&=\sinh x \text{ so } f'(0)=0,\\
f''(x)&=\cosh x \text{ so } f''(0)=1,\\
f^{(3)}(x)&=\sinh x \text{ so } f^{(3)}(0)=0,\\
f^{(4)}(x)&=\cosh x \text{ so } f^{(4)}(0)=1,\\
f^{(5)}(x)&=\sinh x \text{ so } f^{(3)}(0)=0,\\
f^{(6)}(x)&=\cosh x \text{ so } f^{(4)}(0)=1.
\end{aligned}$$

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4+\frac{f^{(5)}(0)}{5!}x^5+\cdots+\frac{f^{(r)}(0)}{r!}x^r.$$

$$\cosh x=1+0+\frac{1}{2!}x^2+0+\frac{1}{4!}x^4+0+\frac{1}{6!}x^6+\cdots$$

$$\therefore\quad\cosh x\approx 1+\frac{1}{2}x^2+\frac{1}{24}x^4.$$

21 b)

$$\lim_{x\to 0}\frac{\cosh x-\cos x}{x^2}=\lim_{x\to 0}\frac{1+\frac{1}{2}x^2+\frac{1}{24}x^4+\cdots-(1-\frac{1}{2}x^2+\frac{1}{24}x^4-\cdots)}{x^2}=1.$$

22 a)

$$(1+y)^{-1}=1-y+y^2-\cdots.$$

22 b)

$$\begin{aligned}
\operatorname{sech}x=\frac{1}{\cosh x}&=\frac{1}{1+(\cosh x-1)}\\
&\approx\frac{1}{1+(\frac{1}{2}x^2+\frac{1}{24}x^4-1)}\\
&\approx 1-\left(\frac{1}{2}x^2+\frac{1}{24}x^4\right)+\left(\frac{1}{2}x^2+\frac{1}{24}x^4\right)^2\\
&\approx 1-\frac{1}{2}x^2-\frac{1}{24}x^4+\frac{1}{4}x^4+\cdots\\
&\approx 1-\frac{1}{2}x^2+\frac{5}{24}x^4.
\end{aligned}$$

22 c)

$$\begin{aligned} f(x) &= \sinh x, & f(0) &= 0, \\ f'(x) &= \cosh x, & f'(0) &= 1, \\ f''(x) &= \sinh x, & f''(0) &= 0, \\ f^{(3)}(x) &= \cosh x, & f^{(3)}(0) &= 1. \end{aligned}$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$\sinh x = 0 + x + 0 + \frac{1}{3!}x^3 + \cdots, \quad \therefore \quad \sinh x \approx x + \frac{1}{6}x^3.$$

22 d)

$$\lim_{x\to 0} \frac{x \sinh 3x}{\operatorname{sech} x - 1} \approx \lim_{x\to 0} \frac{x(3x + \frac{9}{2}x^3)}{(1 - \frac{1}{2}x^2 + \frac{5}{24}x^4) - 1} = \lim_{x\to 0} \frac{3x^2 + \frac{9}{2}x^4}{-\frac{1}{2}x^2 + \frac{5}{24}x^4} = -6.$$

23 a)

$$\begin{aligned} f(x) &= e^{\sin x}, & f(0) &= 1, \\ f'(x) &= \cos x e^{\sin x}, & f'(0) &= 1, \\ f''(x) &= \cos^2 x e^{\sin x} - \sin x e^{\sin x}, & f''(0) &= 1. \end{aligned}$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$e^{\sin x} = 1 + x + \frac{1}{2!}x^2 + \cdots, \quad \therefore \quad e^{\sin x} \approx 1 + x + \frac{1}{2}x^2.$$

23 b)

$$\lim_{x\to 0} \frac{e^{\sin x} - 1}{2x \cos x} \approx \lim_{x\to 0} \frac{x + \frac{1}{2}x^2}{2x - x^3} = \lim_{x\to 0} \frac{1 + \frac{1}{2}x}{2 - x^2} = \frac{1}{2}.$$

24 a)

$$\ln(1 + 2x) = 2x - \frac{1}{2}(2x)^2 + \cdots \approx 2x - 2x^2.$$

24 b)

$$\begin{aligned} f(x) &= \ln(\cos x), & f(0) &= 0, \\ f'(x) &= -\frac{\sin x}{\cos x}, & f'(0) &= 0, \\ f''(x) &= -\frac{\cos^2 x + \sin^2 x}{\cos^2 x} = -\frac{1}{\cos^2 x}, & f''(0) &= -1. \end{aligned}$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots + \frac{f^{(r)}(0)}{r!}x^r.$$

$$\ln(\cos x) = 0 + 0 - \frac{1}{2!}x^2 + \cdots, \quad \therefore \quad \ln(\cos x) \approx -\frac{1}{2}x^2.$$

24 c)

$$\lim_{x\to 0}\frac{x\ln(1+2x)}{\ln(\cos x)-x^2}\approx\lim_{x\to 0}\frac{x(2x-2x^2)}{-\frac{1}{2}x^2-x^2}=\lim_{x\to 0}\frac{2x^2-2x^3}{-\frac{3}{2}x^2}=-\frac{4}{3}.$$

25 a)

$$f(x)=\ln(1+2\sin x),\quad f(0)=0,\quad f'(x)=\frac{2\cos x}{1+2\sin x},\quad f'(0)=2,$$

$$f''(x)=\frac{-2\sin x(1+2\sin x)-4\cos^2 x}{(1+2\sin x)^2},\quad f''(0)=-4.$$

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4+\frac{f^{(5)}(0)}{5!}x^5+\cdots+\frac{f^{(r)}(0)}{r!}x^r.$$

$$\ln(1+2\sin x)=0+2x-\frac{4}{2!}x^2+\cdots,\quad\therefore\quad\ln(1+2\sin x)\approx 2x-2x^2.$$

25 b)

$$\lim_{x\to 0}\frac{\ln(1+2\sin x)}{\ln(1-x)}\approx\lim_{x\to 0}\frac{2x-2x^2}{-x-\frac{1}{2}x^2}=\lim_{x\to 0}\frac{2-2x}{-1-\frac{1}{2}x}=-2.$$

26 a)

$$f(x)=\ln(2-e^x),\quad f(0)=0,\quad f'(x)=-\frac{e^x}{2-e^x},\quad f'(0)=-1,$$

$$f''(x)=\frac{-e^x(2-e^x)-e^{2x}}{(2-e^x)^2},\quad f''(0)=-2.$$

$$f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f^{(3)}(0)}{3!}x^3+\frac{f^{(4)}(0)}{4!}x^4+\frac{f^{(5)}(0)}{5!}x^5+\cdots+\frac{f^{(r)}(0)}{r!}x^r.$$

$$\ln(2-e^x)=0-x-\frac{2}{2!}x^2+\cdots,\quad\therefore\quad\ln(2-e^x)\approx -x-x^2.$$

26 b)

$$\lim_{x\to 0}\frac{x\ln(2-e^x)}{1-\cos kx}\approx\lim_{x\to 0}\frac{x(-x-x^2)}{1-(1-\frac{k^2}{2}x^2)}=-\frac{2}{k^2}.$$

So $-\frac{2}{k^2}=-\frac{1}{8}$ gives $k=\pm 4$.

27)

Substitute $x = \frac{\pi}{4} + t$: $\frac{dx}{dt} = 1$, when $x = \frac{\pi}{4}$, $t = 0$, and when $x = \frac{\pi}{4} + h$, $t = h$.

$$\int_{\pi/4}^{\pi/4+h} \frac{3\sqrt[3]{\cos x}}{2h}\,dx = \int_0^h \frac{3\left(\cos(\frac{\pi}{4}+t)\right)^{1/3}}{2h}\,dt.$$

Let $f(t) = \left(\cos(\frac{\pi}{4}+t)\right)^{1/3}$, so $f(0) = \left(\frac{\sqrt{2}}{2}\right)^{1/3}$. Therefore $f(t) = \left(\frac{\sqrt{2}}{2}\right)^{1/3} + \cdots$. As $h \to 0$, $f(t) \approx \left(\frac{\sqrt{2}}{2}\right)^{1/3}$, so

$$\begin{aligned}\int_0^h \frac{3\left(\cos(\frac{\pi}{4}+t)\right)^{1/3}}{2h}\,dt &\approx \int_0^h \frac{3}{2h}\left(\frac{\sqrt{2}}{2}\right)^{1/3} dt \\ &= \left[\frac{3}{2h}\left(\frac{\sqrt{2}}{2}\right)^{1/3} t\right]_0^h = 3\times 2^{-7/6}.\end{aligned}$$

$$\therefore \quad \lim_{h\to 0}\int_{\pi/4}^{\pi/4+h} \frac{3\sqrt[3]{\cos x}}{2h}\,dx = \frac{3}{2^{7/6}}.$$